

College of Industrial Technology

Seat No.

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1

Year: 2025

Subject: 031001123 Chemistry III

Section: 15-17

Date: Wednesday, October 1, 2025

Time: 10.00 – 12.00

Examination Setter Name: Assoc. Prof. Dr. Sukanya Thepwatee, Asst. Prof. Dr. Jarukorn Sripradite,  
Asst. Prof. Dr. Nisalak Trongsirawat

StudentName.....ID.....Class.....No.....

**Directions:**

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination is divided into 3 sections. There will be 8 pages (including the cover page), 9 questions and they are worth 90 marks (40 points).
3. Complete all questions in this examination paper.
4. Books, documents and lecture notes are not allowed.
5. Electronic calculators are not allowed.
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency.
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

**Cheating in examinations is considered a serious offense and carries the  
penalty of expulsion from student status.**

This exam paper is part of a set used to assess student abilities as follows:

|                                                                               |                                                                                  |                        |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------|
| This exam paper is part of a set used to assess student abilities as follows: |                                                                                  | Question (Q)           |
| CLO 1                                                                         | Explanation and comparison between covalent bond, ionic bond, and metallic bond. |                        |
| CLO 2                                                                         | Description of theoretical term and calculation of chemical bond energies.       | Q3, Q7, Q9             |
| CLO 3                                                                         | Application of fundamental chemistry in other subjects.                          | Q1, Q2, Q4, Q5, Q6, Q8 |
| CLO 4                                                                         | Realization of discipline, punctuality, and public and self-responsibility.      |                        |

Part1 : Total score 30 marks

1. Complete the table below. Please convert the units of the following reaction rates and explain your calculations. (6 marks)

| Substance                      | Reaction rate |          |                                |
|--------------------------------|---------------|----------|--------------------------------|
|                                | in (mol/s)    | in (g/s) | in (dm <sup>3</sup> /s) at STP |
| Ozone (O <sub>3</sub> )        | 0.15          |          |                                |
| Ammonia (NH <sub>3</sub> )     |               |          | 0.224                          |
| Bromine gas (Br <sub>2</sub> ) |               | 56.0     |                                |

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2. Ammonia ( $\text{NH}_3$ ) reacts with oxygen gas ( $\text{O}_2$ ) to produce nitrogen monoxide ( $\text{NO}$ ) and water. In an experiment, the volume of ammonia decreases from  $1,000 \text{ dm}^3$  to  $328 \text{ dm}^3$  at STP within 2 minutes. Please answer the following questions. **(12 marks)**

2.1) Write and balance the chemical equation.

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2.2) Write the rate expression.

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2.3) Determine the consumption rate of ammonia (in  $\text{mol/s}$ ).

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2.4) Determine the consumption rate of oxygen gas (in  $\text{g/s}$ ).

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2.5) Determine the formation rate of nitrogen monoxide (in  $\text{g/s}$ ).

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2.6) Determine the formation rate of water (in  $\text{dm}^3/\text{s}$ ).

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3. Consider the following chemical equation:  $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \rightarrow 2\text{HF}(\text{g})$ . Discuss the factors affecting the reaction rate and explain your reason: **(12 marks)**

3.1) The concentration of hydrogen gas is increased.

- The reaction rate is (Increased or decreased) .....

- Reason: .....

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3.2) A catalyst is added to the reaction.

- The reaction rate is (Increased or decreased) .....

- Reason: .....

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3.3) A noble gas is added.

- The reaction rate is (Increased or decreased) .....

- Reason: .....

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3.4) The temperature of the reaction is decreased.

- The reaction rate is (Increased or decreased) .....

- Reason: .....

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Part 2 : Total score 30 marks

4. The following data were obtained for the reaction: (10 marks)



| Experiment | [A] (M) | [B] (M) | Rate (M/s) |
|------------|---------|---------|------------|
| 1          | 0.10    | 0.10    | 0.020      |
| 2          | 0.20    | 0.10    | 0.040      |
| 3          | 0.20    | 0.20    | 0.080      |

4.1) Determine the order of the reaction with respect to **A** and **B**.

4.2) Write the rate law expression.

4.3) Calculate the rate constant  $k$  with units.

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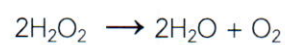
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[illegible]

5. For the decomposition of hydrogen peroxide: (10 marks)



The reaction is first-order in  $[\text{H}_2\text{O}_2]$  with a rate constant of  $k = 0.040 \text{ min}^{-1}$ .

- 5.1) Write the rate law expression.
- 5.2) What is the rate of reaction when  $[\text{H}_2\text{O}_2] = 0.50 \text{ M}$ ?
- 5.3) How long will it take for the concentration of  $[\text{H}_2\text{O}_2]$  to decrease from  $0.50 \text{ M}$  to  $0.10 \text{ M}$ ? (Use:  $\ln(1) = 0$ ,  $\ln(3) = 1.1$ ,  $\ln(5) = 1.6$ )

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6. The decomposition of  $\text{N}_2\text{O}_5$  follows first-order kinetics with a rate constant  $k = 0.1386 \text{ min}^{-1}$

(10 marks)

6.1) Calculate the half-life of the reaction.

6.2) If the initial concentration of  $\text{N}_2\text{O}_5$  is 0.80 M, what will the concentration be after 2 minutes? (Use:  $e^{-0.1386} = 0.8706$  and  $e^{-0.2772} = 0.758$ )

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Part 3 : Total score 30 marks

7. Mark each statement as **True** or **False** and correct the false statements. (7 marks)

7.1) Collision theory accounts for both energy and orientation of molecule.

7.2) Catalysts provide an alternative pathway with lower activation energy.

7.3) In an endothermic reaction, the products have lower energy than the reactants.



7.4) The rate constant,  $k$ , will change when catalyst added.

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7.5) The rate law for an elementary reaction  $A + B \rightarrow C$  is always second order.

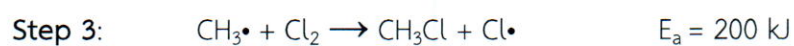
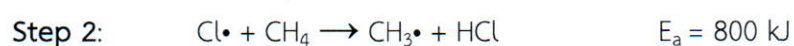
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8. Consider this mechanism:

(16 marks)



8.1) Identify all intermediates.

8.2) What is the overall reaction?

8.3) Determine rate of reaction.

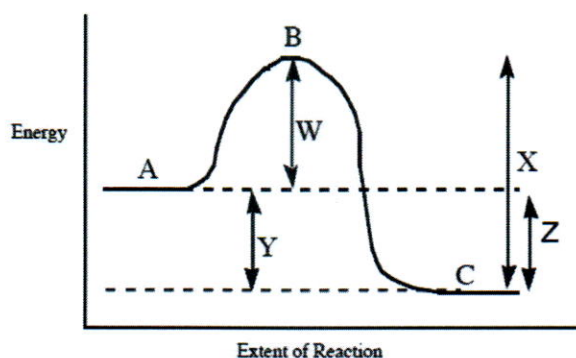
8.4) Which is the rate-determining step?

8.5) How does the radical  $\text{Cl}\cdot$  behave in this mechanism?

8.6) Suggest 2 ways to increase amount of product,  $\text{CH}_3\text{Cl}$ .

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9. For the reaction diagram shown,



9.1) What is  $\Delta H$  for the overall reaction?

9.2) What is  $E_a$  for the reverse reaction?

9.3) Is the overall reaction endothermic or exothermic?

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# PERIODIC TABLE OF ELEMENTS

|                                       |                                        |                                                                                                                                                                                                                                                                           |                                        |                                            |                                        |                                         |                                        |                                      |                                         |                                           |                                          |                                          |                                        |                                        |                                        |                                          |                                         |                                        |                                   |                                      |                                    |
|---------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------|----------------------------------------|-----------------------------------------|----------------------------------------|--------------------------------------|-----------------------------------------|-------------------------------------------|------------------------------------------|------------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|-----------------------------------------|----------------------------------------|-----------------------------------|--------------------------------------|------------------------------------|
| 1                                     | 2                                      | 3                                                                                                                                                                                                                                                                         | 4                                      | 5                                          | 6                                      | 7                                       | 8                                      | 9                                    | 10                                      | 11                                        | 12                                       | 13                                       | 14                                     | 15                                     | 16                                     | 17                                       | 18                                      |                                        |                                   |                                      |                                    |
| 1<br><b>H</b><br>Hydrogen<br>1.008    | Atomic #<br>Symbol<br>Name<br>Weight   |                                                                                                                                                                                                                                                                           |                                        |                                            |                                        |                                         |                                        |                                      |                                         |                                           |                                          |                                          |                                        |                                        |                                        |                                          | 2<br><b>He</b><br>Helium<br>4.0026      |                                        |                                   |                                      |                                    |
| 3<br><b>Li</b><br>Lithium<br>6.94     | 4<br><b>Be</b><br>Beryllium<br>9.0122  | <div><div>C Solid</div><div>Hg Liquid</div><div>H Gas</div><div>Rf Unknown</div></div>                                                                                                                                                                                    |                                        |                                            |                                        |                                         |                                        |                                      |                                         |                                           |                                          |                                          |                                        |                                        |                                        | 5<br><b>B</b><br>Boron<br>10.81          | 6<br><b>C</b><br>Carbon<br>12.011       | 7<br><b>N</b><br>Nitrogen<br>14.007    | 8<br><b>O</b><br>Oxygen<br>15.999 | 9<br><b>F</b><br>Fluorine<br>18.998  | 10<br><b>Ne</b><br>Neon<br>20.180  |
| 11<br><b>Na</b><br>Sodium<br>22.990   | 12<br><b>Mg</b><br>Magnesium<br>24.305 | <div><div>Alkali metals</div><div>Alkaline earth metals</div><div>Lanthanoids (Lanthanides)</div><div>Actinoids (Actinides)</div><div>Transition metals</div><div>Post-transition metals</div><div>Metalloids</div><div>Other nonmetals</div><div>Noble gases</div></div> |                                        |                                            |                                        |                                         |                                        |                                      |                                         |                                           |                                          |                                          |                                        |                                        |                                        | 13<br><b>Al</b><br>Aluminum<br>26.982    | 14<br><b>Si</b><br>Silicon<br>28.085    | 15<br><b>P</b><br>Phosphorus<br>30.974 | 16<br><b>S</b><br>Sulfur<br>32.06 | 17<br><b>Cl</b><br>Chlorine<br>35.45 | 18<br><b>Ar</b><br>Argon<br>39.948 |
| 19<br><b>K</b><br>Potassium<br>39.098 | 20<br><b>Ca</b><br>Calcium<br>40.078   | 21<br><b>Sc</b><br>Scandium<br>44.956                                                                                                                                                                                                                                     | 22<br><b>Ti</b><br>Titanium<br>47.867  | 23<br><b>V</b><br>Vanadium<br>50.942       | 24<br><b>Cr</b><br>Chromium<br>51.996  | 25<br><b>Mn</b><br>Manganese<br>54.938  | 26<br><b>Fe</b><br>Iron<br>55.845      | 27<br><b>Co</b><br>Cobalt<br>58.933  | 28<br><b>Ni</b><br>Nickel<br>58.693     | 29<br><b>Cu</b><br>Copper<br>63.546       | 30<br><b>Zn</b><br>Zinc<br>65.38         | 31<br><b>Ga</b><br>Gallium<br>69.723     | 32<br><b>Ge</b><br>Germanium<br>72.630 | 33<br><b>As</b><br>Arsenic<br>74.922   | 34<br><b>Se</b><br>Selenium<br>78.971  | 35<br><b>Br</b><br>Bromine<br>79.904     | 36<br><b>Kr</b><br>Krypton<br>83.798    |                                        |                                   |                                      |                                    |
| 37<br><b>Rb</b><br>Rubidium<br>85.468 | 38<br><b>Sr</b><br>Strontium<br>87.62  | 39<br><b>Y</b><br>Yttrium<br>88.906                                                                                                                                                                                                                                       | 40<br><b>Zr</b><br>Zirconium<br>91.224 | 41<br><b>Nb</b><br>Niobium<br>92.906       | 42<br><b>Mo</b><br>Molybdenum<br>95.95 | 43<br><b>Tc</b><br>Technetium<br>(98)   | 44<br><b>Ru</b><br>Ruthenium<br>101.07 | 45<br><b>Rh</b><br>Rhodium<br>102.91 | 46<br><b>Pd</b><br>Palladium<br>106.42  | 47<br><b>Ag</b><br>Silver<br>107.87       | 48<br><b>Cd</b><br>Cadmium<br>112.41     | 49<br><b>In</b><br>Indium<br>114.82      | 50<br><b>Sn</b><br>Tin<br>118.71       | 51<br><b>Sb</b><br>Antimony<br>121.76  | 52<br><b>Te</b><br>Tellurium<br>127.60 | 53<br><b>I</b><br>Iodine<br>126.90       | 54<br><b>Xe</b><br>Xenon<br>131.29      |                                        |                                   |                                      |                                    |
| 55<br><b>Cs</b><br>Caesium<br>132.91  | 56<br><b>Ba</b><br>Barium<br>137.33    | 57–71                                                                                                                                                                                                                                                                     |                                        | 72<br><b>Hf</b><br>Hafnium<br>178.49       | 73<br><b>Ta</b><br>Tantalum<br>180.95  | 74<br><b>W</b><br>Tungsten<br>183.84    | 75<br><b>Re</b><br>Rhenium<br>186.21   | 76<br><b>Os</b><br>Osmium<br>190.23  | 77<br><b>Ir</b><br>Iridium<br>192.22    | 78<br><b>Pt</b><br>Platinum<br>195.08     | 79<br><b>Au</b><br>Gold<br>196.97        | 80<br><b>Hg</b><br>Mercury<br>200.59     | 81<br><b>Tl</b><br>Thallium<br>204.38  | 82<br><b>Pb</b><br>Lead<br>207.2       | 83<br><b>Bi</b><br>Bismuth<br>208.98   | 84<br><b>Po</b><br>Polonium<br>(209)     | 85<br><b>At</b><br>Astatine<br>(210)    | 86<br><b>Rn</b><br>Radon<br>(222)      |                                   |                                      |                                    |
| 87<br><b>Fr</b><br>Francium<br>(223)  | 88<br><b>Ra</b><br>Radium<br>(226)     | 89–103                                                                                                                                                                                                                                                                    |                                        | 104<br><b>Rf</b><br>Rutherfordium<br>(267) | 105<br><b>Db</b><br>Dubnium<br>(268)   | 106<br><b>Sg</b><br>Seaborgium<br>(269) | 107<br><b>Bh</b><br>Bohrium<br>(270)   | 108<br><b>Hs</b><br>Hassium<br>(277) | 109<br><b>Mt</b><br>Meitnerium<br>(278) | 110<br><b>Ds</b><br>Darmstadtium<br>(281) | 111<br><b>Rg</b><br>Roentgenium<br>(282) | 112<br><b>Cn</b><br>Copernicium<br>(285) | 113<br><b>Nh</b><br>Nihonium<br>(286)  | 114<br><b>Fl</b><br>Flerovium<br>(289) | 115<br><b>Mc</b><br>Moscovium<br>(290) | 116<br><b>Lv</b><br>Livermorium<br>(293) | 117<br><b>Ts</b><br>Tennessine<br>(294) | 118<br><b>Og</b><br>Ognesson<br>(294)  |                                   |                                      |                                    |